

Lecture 5: Solving Linear Equations and Null Space

MATH203 Applied Linear Algebra

Motivation: Why We Need to Solve Equations

Recall: The Two Languages (Lecture 4)

Every subspace has **two complementary descriptions**:

Language	How it works	Example
Descriptive	List equations (constraints)	$\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
Constructive	Provide recipe (generate members)	$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots\}$

Today's question: What is each language **good at**? What is each language **bad at**?

Each Language Has Strengths and Weaknesses

	Descriptive $\{x : Ax = 0\}$	Constructive $\text{span}\{v_1, \dots\}$
Good at	Verification (check membership)	Generation (produce members)
Bad at	Generation (find examples)	Verification (check membership)

Key insight: Solving equations helps both languages do what they're bad at!

Let's see two concrete examples using Shinnan's coffee shop...

Example Setup: Shinchin's Coffee Shop Menu

Shinchin has three drinks on his menu:

	 milk	 coffee	 tea
 tea leaves	0	0	2
 lemon	0	0	1
 beans	0	2	4
 milk	1	0	1

Matrix form:

$$A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$

Case 1: Column Space (Naturally Constructive)

The column space is **naturally constructive**:

$$\text{Col}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} \right\}$$

These are the ingredient lists for , , .

In coffee shop language: $\text{Col}(A)$ = all possible ingredient combinations we can produce from our menu.

Case 1: Generation is Easy

Generation (easy): Want a vector in $\text{Col}(A)$?

Just pick a combination of drinks!

$$2 \text{ 🥛 } + 3 \text{ ☕ } = 2 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 6 \\ 2 \end{pmatrix}$$

In coffee shop language: "Make 2 glasses of milk and 3 cups of coffee" produces an ingredient combination in $\text{Col}(A)$.

Generation = **making products**. Easy with constructive description!

Case 1: Verification is Hard

Verification (hard): A customer brings an ingredient list $\mathbf{b} = \begin{pmatrix} 4 \\ 2 \\ 10 \\ 3 \end{pmatrix}$.

Question in coffee shop language: Can we make a drink combination that uses exactly these ingredients?

This is asking: Is $\mathbf{b} \in \text{Col}(A)$?

Can't tell by looking! Need to **solve**: $A\mathbf{x} = \mathbf{b}$

- If solution exists $\Rightarrow$ Yes, we can make it! ($\mathbf{b} \in \text{Col}(A)$)
- If no solution $\Rightarrow$ No, impossible with our menu ($\mathbf{b} \notin \text{Col}(A)$)

Solving equations gives constructive description the power to verify!

Case 2: Null Space (Naturally Descriptive)

The null space is **naturally descriptive**:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^3 : A\mathbf{x} = \mathbf{0}\}$$

In coffee shop language: $\text{Null}(A) =$ **drink combinations that produce zero net ingredients.**

This happens when drinks are redundant: some combination cancels out!

For example: If  = 2 × , then 2  - 1  = **0** ingredients.

Geometric meaning: $\text{Null}(A)$ records **relationships between products.**

Case 2: Verification is Easy (Coffee Shop Example)

Verification (easy): Someone claims "1 milk + 1 coffee - 2 tea = zero ingredients."

$$\text{Is } \mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \text{ in Null}(A)?$$

Just compute in coffee shop:

$$\begin{aligned} 1 \text{ 🥛} + 1 \text{ ☕} - 2 \text{ 🍵} &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -4 \\ -2 \\ -6 \\ -1 \end{pmatrix} \neq \mathbf{0} \end{aligned}$$

Coffee shop conclusion: This combination does NOT produce zero!

Case 2: Generation is Hard

Generation (hard): Looking at the menu, can you find the redundancy?

			
	0	0	2
	0	0	1
	0	2	4
	1	0	1

Question in coffee shop language: Which drinks are redundant?
What relationship do they satisfy?

Hard to see by staring! Need to **solve** $Ax = 0$.

Case 2: Solving Reveals the Relationship

After solving $A\mathbf{x} = \mathbf{0}$ (we'll learn how soon):

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \right\}$$

What does this tell us in coffee shop language?

$$0 \cdot \text{milkshake} + 2 \cdot \text{coffee} - 1 \cdot \text{matcha} = 0$$

Rearranging: $2 \times \text{coffee} = 1 \times \text{matcha}$

Aha! The  recipe uses exactly **twice the ingredients** of . That's the redundancy!

Solving equations gives descriptive description the power to generate!

Summary: The Power of Solving Equations

Subspace	Weakness	Solving provides
$\text{Col}(A)$ (constructive)	Verification "Can we make this?"	Solve $A\mathbf{x} = \mathbf{b}$ to check if possible
$\text{Null}(A)$ (descriptive)	Generation "What relationships?"	Solve $A\mathbf{x} = \mathbf{0}$ to find relationships

Today's focus: Develop a **cross-filling method** to solve $A\mathbf{x} = \mathbf{b}$, which:

- Finds all relationships between products (basis for $\text{Null}(A)$)
- Checks if a product is possible ($\mathbf{b} \in \text{Col}(A)$?)
- Determines uniqueness (Is there only one way?)

Null Space: Starting with the Descriptive Definition

Definition: Null Space

Definition (Null Space (Kernel))

For a matrix $A \in \mathbb{R}^{m \times n}$, the **null space** is:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Meaning: $\text{Null}(A)$ stores all **relationships between the columns** of A .

Also called the **kernel**, denoted $\ker(A)$.

In coffee shop language: Drink combinations producing zero net ingredients.

Immediate Example: Finding Relationships

From Shinchin's menu:

			
	0	0	2
	0	0	1
	0	2	4
	1	0	1

Question: Are there relationships between these drinks?

If we find $\mathbf{x} \in \text{Null}(A)$, it reveals the relationship!

What the Null Space Vector Tells Us

Suppose we found:

$$\mathbf{x} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \in \text{Null}(A)$$

Column view of $A\mathbf{x} = \mathbf{0}$:

$$0 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} + (-1) \cdot \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Rearranging:

$$2 \cdot \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix}$$

Key Insight: Null Space as Relationship Catalog

Important:

- $\mathbf{x} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ doesn't mean "2 coffees minus 1 tea" (nonsense!)
- It records:  = 2 ×  (a relationship between products)
- $\text{Null}(A) = \text{"relationship catalog"}$

Size of the catalog:

- $\text{Null}(A) = \{\mathbf{0}\} \rightarrow$ no relationships $\rightarrow$ columns **independent**
- $\dim(\text{Null}(A)) = k \rightarrow$ there are k independent relationships

Null Space is a Subspace

Proposition

$\text{Null}(A)$ is a subspace of $\mathbb{R}^n$.

Proof: Verify the three axioms:

1. $\mathbf{0} \in \text{Null}(A)$ since $A\mathbf{0} = \mathbf{0}$ ✓
2. If $A\mathbf{x}_1 = \mathbf{0}$ and $A\mathbf{x}_2 = \mathbf{0}$, then

$$A(\mathbf{x}_1 + \mathbf{x}_2) = \mathbf{0}$$

✓

3. If $A\mathbf{x} = \mathbf{0}$, then $A(c\mathbf{x}) = \mathbf{0}$ ✓

The Problem: We Need Constructive Description

Descriptive: $\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$

- **Verification** (easy): Is $\mathbf{v}$ in $\text{Null}(A)$? Compute $A\mathbf{v}$
- **Generation** (hard): Find relationships? Can't guess!

Goal: Convert to **constructive**:

$$\text{Null}(A) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

Method: Solve $A\mathbf{x} = \mathbf{0}$ using cross-filling!

But wait — to solve $A\mathbf{x} = \mathbf{b}$ in general, we need to understand augmented matrices first...

From $Ax = b$ to Augmented Matrix

Motivation: Why Augmented Matrix?

We want to solve: $A\mathbf{x} = \mathbf{b}$

Column view:

$$A\mathbf{x} = \left(\begin{array}{c|c|c|c} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{array} \right) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

We're looking for **coefficients** $x_1, \dots, x_n$ that combine columns to get $\mathbf{b}$.

Key idea: Add $\mathbf{b}$ as an extra column, then find $\mathbf{x}$ such that this augmented system satisfies a special property...

The Augmented Matrix

Form the **augmented matrix**:

$$(A \mid \mathbf{b}) = \left(\begin{array}{c|c|ccc|c} & & & & & \\ & & & & & \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \\ & & & & \\ & & & & \end{array} \right)$$

This is an $m \times (n + 1)$ matrix with $n + 1$ columns.

Now consider: What happens if we take a linear combination of these

columns with coefficients $\begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix}$?

The Key Connection

Take the augmented matrix $(A \mid \mathbf{b})$ and multiply by $\begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix}$:

$$(A \mid \mathbf{b}) \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix} = x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n + (-1)\mathbf{b}$$

This equals $\mathbf{0}$ if and only if:

$$x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

which is exactly $A\mathbf{x} = \mathbf{b}$!

Why This Matters

Crucial observation:

$$A\mathbf{x} = \mathbf{b} \text{ has solution} \iff \begin{pmatrix} \mathbf{x} \\ -1 \end{pmatrix} \in \text{Null}(A \mid \mathbf{b})$$

In other words:

- Solving $A\mathbf{x} = \mathbf{b}$ = finding null space vectors of $(A \mid \mathbf{b})$ with **last entry = -1**
- The augmented matrix's null space contains all the information about solutions!

But actually, we won't compute $\text{Null}(A \mid \mathbf{b})$ directly. Instead, we'll use **cross-filling** on $(A \mid \mathbf{b})$ to extract equations...

The Cross-Filling Method for Solving $Ax = b$

Recall: Cross-Filling Decomposition (Lecture 3)

Any matrix can be decomposed via cross-filling:

$$A = R_1 + R_2 + \cdots + R_r$$

where each R_i is a **rank-one matrix**.

Key property of rank-one matrices: All rows are scalar multiples of each other.

Example:

$$R = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{pmatrix}$$

Row 1 = 2 × Row 2, Row 3 = 3 × Row 2

Apply Cross-Filling to Augmented Matrix

For $(A \mid \mathbf{b})$, we can cross-fill:

$$(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + (R_2 \mid \mathbf{b}_2) + \cdots + (R_r \mid \mathbf{b}_r)$$

Each piece $(R_i \mid \mathbf{b}_i)$ is a rank-one $(m \times (n + 1))$ matrix.

Key observation: Since $(R_i \mid \mathbf{b}_i)$ is rank-one, all its rows are proportional — they represent the **same equation!**

Example: Rank-One = One Equation

Consider:

$$(R_1 \mid \mathbf{b}_1) = \left(\begin{array}{ccc|c} 2 & 4 & 6 & 8 \\ 1 & 2 & 3 & 4 \\ 3 & 6 & 9 & 12 \end{array} \right)$$

All three rows give the same equation:

- Row 1: $2x_1 + 4x_2 + 6x_3 = 8 \div 2 \rightarrow x_1 + 2x_2 + 3x_3 = 4$
- Row 2: $x_1 + 2x_2 + 3x_3 = 4$
- Row 3: $3x_1 + 6x_2 + 9x_3 = 12 \div 3 \rightarrow x_1 + 2x_2 + 3x_3 = 4$

Same equation! One rank-one piece = one independent equation.

Why Are These Equations Valid?

Question: We extract one equation from each rank-one piece. Why is solving these r equations equivalent to the original m equations?

Answer: Cross-filling gives $(A \mid \mathbf{b}) = UV$ where:

- U is $m \times r$ with **linearly independent columns**
- V is $r \times (n + 1)$ with rows from each piece

Since U 's columns are independent, we can **left-cancel**:

$$UV\mathbf{z} = \mathbf{0} \iff V\mathbf{z} = \mathbf{0}$$

Therefore: Solutions to $(A \mid \mathbf{b}) =$ Solutions to V

What is V ?

$$V = \begin{pmatrix} \text{— row from } (R_1 \mid \mathbf{b}_1) \text{ —} \\ \text{— row from } (R_2 \mid \mathbf{b}_2) \text{ —} \\ \vdots \\ \text{— row from } (R_r \mid \mathbf{b}_r) \text{ —} \end{pmatrix}$$

We've reduced m equations to r equations ($r \leq m$).

Each row of V is one simplified equation from cross-filling.

Observation: Variables Decrease

As cross-filling proceeds, each piece has fewer active variables:

- $(R_1 \mid \mathbf{b}_1)$: involves **all n variables**
- $(R_2 \mid \mathbf{b}_2)$: involves **$n - 1$ variables** (column j_1 eliminated)
- $(R_3 \mid \mathbf{b}_3)$: involves **$n - 2$ variables**
- $\vdots$
- $(R_r \mid \mathbf{b}_r)$: involves **$n - r + 1$ variables** (fewest!)

Strategy: Start from the end (fewest variables) and work backwards!

Backward Solving Strategy

Step 1: Read $(R_r \mid \mathbf{b}_r)$ (simplest!)

- Solve for its pivot variable

Step 2: Read $(R_{r-1} \mid \mathbf{b}_{r-1})$

- Solve for its pivot variable

Step 3: Continue backwards. . .

Results appear naturally:

- **Particular solution:** Set free variables = 0
- **Null space basis:** Set each free variable = 1 (one at a time)
- **Existence:** Check if any contradiction ($0 = c$, $c \neq 0$)
- **Uniqueness:** Check if free variables exist

Concrete Example: Step-by-Step

Example: Solve $A\mathbf{x} = \mathbf{b}$

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 1 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 11 \\ 16 \end{pmatrix}$$

Form augmented matrix:

$$(A \mid \mathbf{b}) = \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 1 & 2 & 11 \\ 3 & 6 & 1 & 3 & 16 \end{array} \right]$$

Step 1: Cross-Fill — Choose Pivot (1,1)

Pivot at entry $(1,1) = 1$:

$$\begin{aligned} R_1 &= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \end{array} \right) \\ &= \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 0 & 2 & 10 \\ 3 & 6 & 0 & 3 & 15 \end{array} \right] \end{aligned}$$

Equation from R_1 : $x_1 + 2x_2 + 0x_3 + x_4 = 5$

Step 2: Compute Remainder

Subtract R_1 from $(A \mid \mathbf{b})$:

$$\text{Remainder} = \left[\begin{array}{cccc|c} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right]$$

Pivot at entry $(2, 3) = 1$:

$$R_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} (0 \quad 0 \quad 1 \quad 0 \quad | \quad 1)$$

Equation from R_2 : $x_3 = 1$

Step 3: Decomposition Complete

$$(A \mid \mathbf{b}) = R_1 + R_2$$

Two equations:

$$\text{From } R_1 : x_1 + 2x_2 + 0x_3 + x_4 = 5$$

$$\text{From } R_2 : x_3 = 1$$

Pivot columns: 1, 3 $\rightarrow$ **Pivot variables:** x_1, x_3

Non-pivot columns: 2, 4 $\rightarrow$ **Free variables:** x_2, x_4

Existence check: No contradiction ($0 = c$) $\rightarrow$ solution exists!

Step 4: Backward Solving

From R_2 (simplest): $x_3 = 1$

From R_1 : $x_1 + 2x_2 + 0x_3 + x_4 = 5$

Solve for pivot x_1 :

$$x_1 = 5 - 2x_2 - x_4$$

General solution: Let $x_2 = s$, $x_4 = t$ (free parameters)

$$\mathbf{x} = \begin{pmatrix} 5 - 2s - t \\ s \\ 1 \\ t \end{pmatrix}$$

Step 5: Decompose the Solution

$$\mathbf{x} = \begin{pmatrix} 5 - 2s - t \\ s \\ 1 \\ t \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Particular solution ($s = t = 0$): $\mathbf{x}_p = \begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

Null space basis:

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Applications: Existence, Uniqueness, and Null Space

Application 1: Finding Null Space Basis

Proposition

To find a basis for $\text{Null}(A)$:

1. Solve $A\mathbf{x} = \mathbf{0}$ using cross-filling
2. Identify free variables
3. For each free variable, set it = 1, others = 0
4. Solve for pivot variables

Result: $\text{Null}(A) = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$

Descriptive $\rightarrow$ Constructive conversion complete!

Application 2: Existence Criterion

Theorem (Existence of Solutions)

$A\mathbf{x} = \mathbf{b}$ has a solution if and only if:

$$\text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$$

Understanding via cross-filling:

- Choose all pivots in A 's part (first n columns)
- If $\mathbf{b}$ column is completely accounted for $\rightarrow$ rank same $\rightarrow$ solution exists
- If remainder in $\mathbf{b}$ column $\neq 0 \rightarrow$ get equation $0 = c$ ($c \neq 0$) $\rightarrow$ no solution

Geometric Interpretation of Existence

$$\mathbf{b} \in \text{Col}(A) \iff \text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$$

Adding $\mathbf{b}$ as a column doesn't increase rank $\iff \mathbf{b}$ is already in span of A 's columns.

Coffee shop language: The customer's ingredient list is achievable $\iff$ it's in our column space.

Application 3: Uniqueness Criterion

Theorem (Uniqueness of Solutions)

If $A\mathbf{x} = \mathbf{b}$ has a solution, it is **unique** if and only if:

$$\text{Null}(A) = \{\mathbf{0}\}$$

Equivalently: No free variables (all columns are pivot columns).

Also: $\text{rank}(A) = n$ (number of columns).

Proof idea: If $\mathbf{x}_1, \mathbf{x}_2$ are both solutions, then $\mathbf{x}_1 - \mathbf{x}_2 \in \text{Null}(A)$.

Application 4: Structure of Solution Set

Theorem (General Solution Structure)

If $A\mathbf{x} = \mathbf{b}$ has solutions, then **all solutions** have the form:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$$

where:

- $\mathbf{x}_p$ is any **particular solution**
- $\mathbf{x}_n \in \text{Null}(A)$ is arbitrary

Geometric picture: Solution set is $\mathbf{x}_p + \text{Null}(A)$ (affine subspace).

Dimension Formula: Rank-Nullity Theorem

Counting Pivot and Free Variables

From cross-filling an $m \times n$ matrix A :

- Number of pivot columns = $\text{rank}(A)$
- Number of free variables = $\dim(\text{Null}(A))$
- Total columns = n

Every column is either pivot or non-pivot, so:

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

This is the **Rank-Nullity Theorem**.

Rank-Nullity Theorem

Theorem (Rank-Nullity Theorem)

For any $m \times n$ matrix A :

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

Proof idea: Every column is either:

- A pivot column (contributes to rank), or
- A non-pivot column (contributes one free variable to null space basis)

These counts sum to n .

Example: Verifying Rank-Nullity

From our earlier example:

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 1 & 3 \end{pmatrix}$$

- $\text{rank}(A) = 2$ (two pivot columns: 1, 3)
- $\dim(\text{Null}(A)) = 2$ (two free variables: x_2, x_4)
- $n = 4$ (total columns)

Check: $2 + 2 = 4$ ✓

Preview: Null Space and Row Space Orthogonality

Natural Orthogonality

Theorem

$$\text{Null}(A) \perp \text{Row}(A)$$

Every vector in $\text{Null}(A)$ is orthogonal to every vector in $\text{Row}(A)$.

Proof: Let $\mathbf{x} \in \text{Null}(A)$ and $\mathbf{r}_i = i\text{-th row of } A$.

By definition: $A\mathbf{x} = \mathbf{0}$

This means:

$$\begin{pmatrix} - & \mathbf{r}_1 & - \\ & \vdots & \\ - & \mathbf{r}_m & - \end{pmatrix} \mathbf{x} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

So $\mathbf{r}_i \cdot \mathbf{x} = 0$ for all i .

Coffee Shop Interpretation

If $A = \text{"drinks"} \rightarrow \text{ingredients}$:

- **Row space:** All ingredient combinations from drinks
- **Null space:** Drink combinations producing zero ingredients

These are perpendicular: A drink combination producing zero has nothing in common with actual ingredient requirements!

Next lecture: Four fundamental subspaces and complete orthogonal structure.

Summary

What We Achieved Today

1. Null Space:

- Descriptive: $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
- Records relationships between columns
- Conversion to constructive requires solving

2. Cross-Filling Method:

- Each rank-one piece = one equation
- Variables decrease $\rightarrow$ backward solving
- Natural structure: particular + null space

What We Achieved Today (continued)

3. **Existence**: $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) \iff \text{solution exists}$

4. **Uniqueness**: $\text{Null}(A) = \{\mathbf{0}\} \iff \text{solution unique}$

5. **Rank-Nullity**: $\text{rank}(A) + \dim(\text{Null}(A)) = n$

6. **Orthogonality**: $\text{Null}(A) \perp \text{Row}(A)$

Next lecture: Four fundamental subspaces and complete orthogonal decomposition!